

Title

Students' Use of Artificial Intelligence in South African Education: Dependence or Support for Independent Learning?

Introduction

Artificial Intelligence tools such as ChatGPT, intelligent tutoring systems, and AI-powered learning platforms are increasingly used by students to assist with studying, completing assignments, and understanding complex concepts. In South Africa, students across secondary schools and higher education institutions are beginning to rely on AI for academic support, particularly in environments where teacher support and learning resources may be limited.

While AI offers opportunities to enhance learning, there is growing concern that excessive reliance on AI may reduce students' critical thinking, problem-solving abilities, and independent learning skills. This raises an important question: will future students become overly dependent on AI, or will they develop the knowledge and skills necessary to succeed even without AI? This study aims to explore how students in South Africa currently use AI and how this use affects their learning and skill development.

Background and Rationale

South African students face diverse educational challenges, including large class sizes, limited access to textbooks, and unequal access to digital resources. AI tools can help students by providing explanations, practice questions, and instant feedback. As a result, many students view AI as a valuable learning aid.

However, there is concern that students may use AI as a shortcut to complete academic tasks rather than as a tool to support learning. Understanding how students use AI—whether as a learning assistant or as a replacement for thinking—is important for ensuring that education continues to develop essential skills such as reasoning, creativity, and problem-solving.

Research Problem

Although AI tools are increasingly used by students in South Africa, little is known about how this use affects their independence, academic skills, and long-term learning ability. There is uncertainty about whether reliance on AI will weaken students' foundational knowledge and skills or whether AI can be used in ways that support independent learning.

Research Objectives

The objectives of this study are to:

1. Investigate how students in South Africa use AI tools for learning and academic tasks.
2. Examine students' levels of dependence on AI for completing assignments and studying.
3. Explore students' perceptions of whether AI improves or weakens their understanding and skills.
4. Assess whether students believe they could succeed academically without AI support.

Research Questions

1. How do South African students use AI tools in their studies?
2. To what extent do students depend on AI for learning and completing academic work?
3. How does AI use influence students' critical thinking and problem-solving skills?
4. Do students believe they have the knowledge and skills to succeed without AI?

Literature Review

Existing research shows that AI can support learning by offering personalized explanations and immediate feedback. However, some studies suggest that over-reliance on AI may reduce students' ability to think independently and solve problems without assistance. In the South African context, limited research exists on students' AI dependency and its impact on skill development. This study aims to contribute to this area by focusing specifically on students' experiences and perceptions.

Research Methodology

Research Design

This study will use a **qualitative or mixed-methods approach**, suitable for undergraduate research.

Data Collection

- **Questionnaires** distributed to students to gather information on how often and why they use AI tools.
- **Open-ended questions or short interviews** to explore students' views on dependence on AI and independent learning.

Sample

The study will involve undergraduate students or senior secondary school learners from selected South African educational institutions that have access to digital learning tools.

Data Analysis

- Questionnaire responses will be analysed using descriptive statistics (e.g., frequencies and percentages).
- Qualitative responses will be analysed thematically to identify patterns related to AI dependence and skill development.

Ethical Considerations

Ethical approval will be obtained from the relevant institution. Participation will be voluntary, and informed consent will be secured. Students' identities will remain anonymous, and all data will be used strictly for academic purposes.

Significance of the Study

This study will provide insights into how students in South Africa use AI in their learning and whether this use supports or undermines independent knowledge and skills. The findings may help educators and institutions develop guidelines for responsible AI use that encourage learning rather than dependency.

Proposed Timeline

Activity	Duration
Literature Review	1 month
Research Design & Ethics Approval	1 month
Data Collection	2 months
Data Analysis	1 month
Writing and Submission	2 months

Conclusion

AI has become an important part of students' learning experiences in South Africa. This study seeks to examine whether AI serves as a helpful learning tool or creates dependence that may weaken students' knowledge and skills. The findings will contribute to discussions on how AI can be used responsibly to support, rather than replace, independent learning.